

Viewframe PRD

Product Name

Viewframe

Tagline

Figma Prototype Mode for Live Websites

Preview any live website inside realistic device mockups with synchronized responsive breakpoints, orientation switching, and visual review boards.

1. Vision

Designers and developers struggle to review responsive layouts across tablet, laptop, and desktop breakpoints.

Browser DevTools provide accurate rendering but poor presentation.

Mockup generators create static screenshots but cannot display live websites.

Viewframe bridges this gap by creating a visual review workspace where users can inspect live websites across multiple devices simultaneously.

2. Problem Statement

Current workflow problems:

- DevTools only show one viewport at a time.
- Reviewing tablet breakpoints on smaller screens is cumbersome.
- Existing mockup generators are static.
- Most responsive testing tools are overly technical.
- Stakeholders cannot easily review responsive layouts.

Teams need a visual workspace that combines:

- Live websites
- Real viewport rendering
- Device mockups

- Multi-device comparison
 - Responsive QA workflows
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3. Product Goals

Primary Goals

- Allow users to preview any public website.
- Support tablet, laptop, desktop, and custom breakpoints.
- Display realistic device mockups.
- Support portrait and landscape orientations.
- Enable side-by-side responsive comparison.

Secondary Goals

- Improve responsive QA workflows.
 - Create presentation-quality previews.
 - Reduce dependency on browser DevTools.
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4. Target Users

Designers

Review responsive layouts visually.

Frontend Developers

Validate breakpoint implementations.

Product Teams

Review responsive experiences before release.

Agencies

Present websites to clients.

5. Core User Stories

Designer

As a designer, I want to view a website in tablet portrait and landscape simultaneously so I can validate responsive layouts.

Developer

As a developer, I want to compare multiple breakpoints at once so I can identify responsive issues faster.

Agency

As an agency owner, I want to generate presentation-ready mockups from a live website.

Product Manager

As a product manager, I want to review responsive layouts across multiple devices in one view.

6. MVP Features

6.1 URL Input

Users can enter:

`https://example.com`

and instantly generate previews.

6.2 Device Library

Built-in presets:

Tablets

- iPad Mini
- iPad Air
- iPad Pro
- Surface Pro

Laptops

- MacBook Air
- MacBook Pro
- Surface Laptop

Desktops

- Small Desktop (1280×720)
- Desktop HD (1366×768)
- Desktop Full HD (1920×1080)

Custom Device

Users can define:

- Width
 - Height
 - Name
 - Orientation
-

6.3 Orientation Switching

Supported modes:

Portrait

834 × 1194

Landscape

1194 × 834

Viewport dimensions update dynamically.

6.4 Infinite Canvas

Users can:

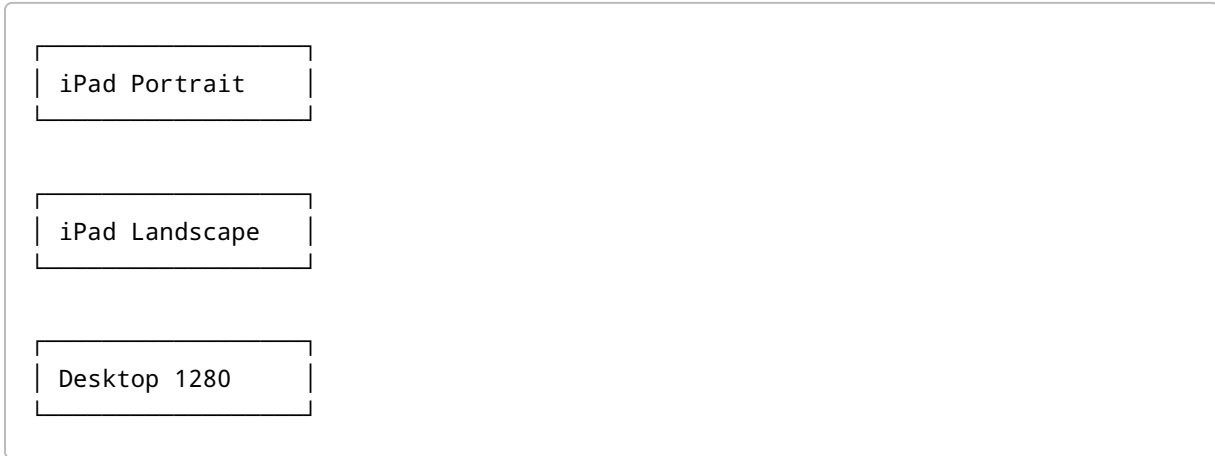
- Pan
- Zoom
- Rearrange devices

- Group devices

Similar to Figma.

6.5 Responsive Board

Example:



All connected to the same website.

6.6 Linked Navigation

When one frame navigates:

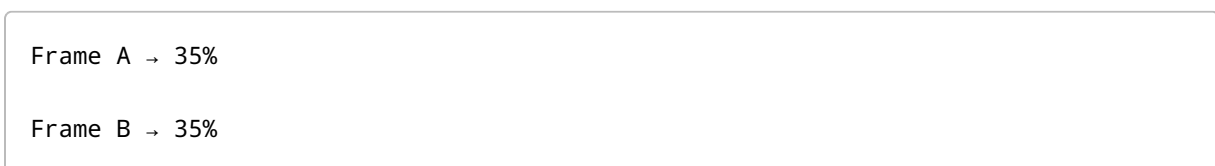


All connected frames update.

6.7 Linked Scroll

Optional synchronized scrolling.

Example:



Frame C → 35%

6.8 Mockups

Modes:

Realistic

Device bezels and hardware.

Minimal

Simple viewport frame.

Wireframe

Border-only mode.

6.9 Scaling Modes

Actual Viewport

Website renders at true viewport dimensions.

Fit-to-Canvas

Entire mockup scales proportionally.

Viewport dimensions remain unchanged internally.

This ensures responsive behavior remains accurate while fitting larger devices into smaller screens.

6.10 Export

Export:

- PNG
- PDF

Supported exports:

- Entire board
 - Individual device
 - Selected group
-

7. Future Features

Collaboration

- Comments
- Annotations
- Multiplayer editing

Browser Testing

- Chrome
- Safari
- Firefox

Responsive Heatmaps

Identify layout shifts.

AI QA Assistant

Detect responsive issues automatically.

Figma Integration

Generate boards directly from designs.

8. Success Metrics

Product Metrics

- Weekly Active Users
 - Boards Created
 - Devices Added Per Board
 - Export Frequency
-

9. Technical Constraints

iframe Limitation

Many public websites block iframe embedding through:

`X-Frame-Options`

and

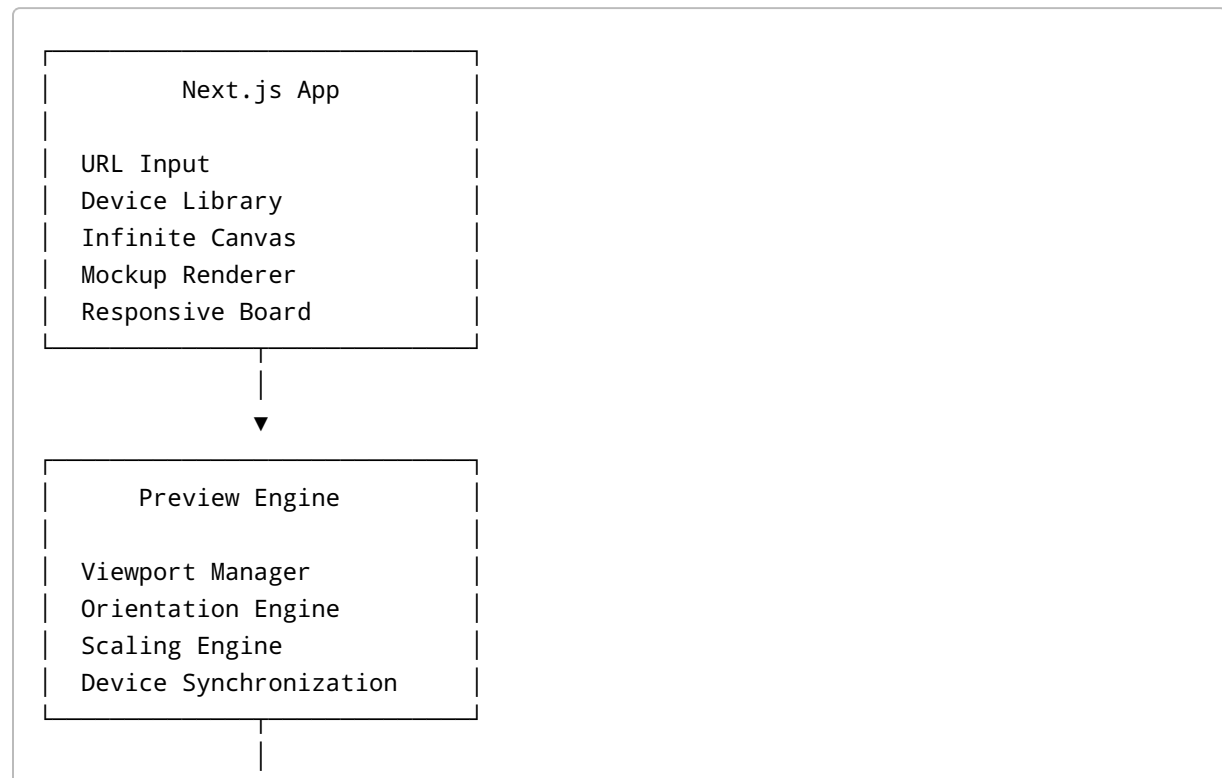
`Content-Security-Policy`

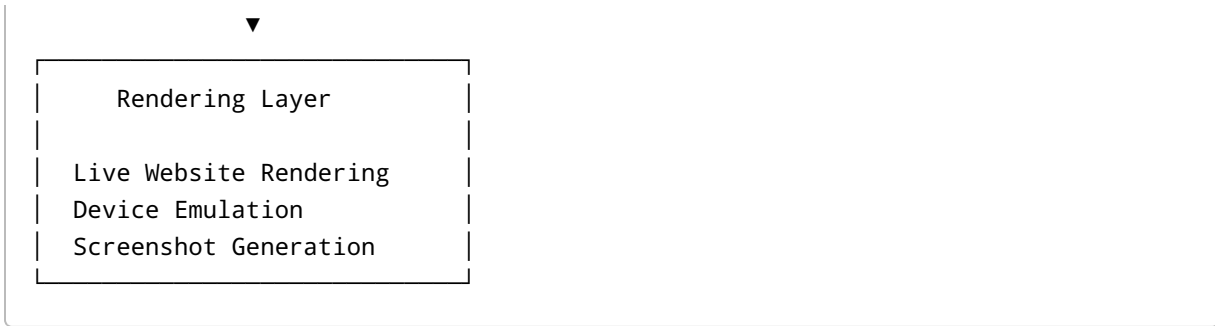
Therefore Viewframe cannot rely entirely on iframe rendering.

The rendering architecture must support websites that restrict embedding.

10. Architecture

High Level Architecture

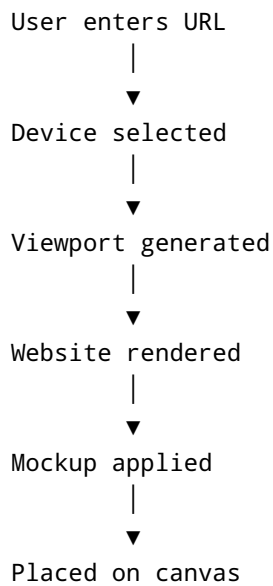




11. Frontend Architecture

```
src
├── app
├── components
│   ├── canvas
│   ├── devices
│   ├── mockups
│   ├── toolbar
│   └── preview
├── features
│   ├── board
│   ├── viewport
│   ├── orientation
│   ├── navigation-sync
│   └── export
├── hooks
├── store
├── lib
└── types
```

12. Rendering Flow



13. Tech Stack

Core Framework

- Next.js 15
- React 19
- TypeScript

Styling

- Tailwind CSS
- shadcn/ui

Canvas Engine

Recommended

tldraw SDK

Benefits:

- Infinite canvas
 - Pan
 - Zoom
 - Selection
 - Dragging
 - Future collaboration support
-

State Management

Zustand

Used for:

- Device state
 - Canvas state
 - Orientation state
 - Synchronization state
-

Animation

Framer Motion

Used for:

- Orientation transitions
 - Device placement
 - Canvas interactions
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Device Mockups

SVG-based device frames.

Benefits:

- Lightweight
 - Scalable
 - Resolution independent
 - Easy orientation switching
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Rendering Engine

Playwright

Responsibilities:

- Website rendering
 - Device emulation
 - Accurate viewport simulation
 - Screenshot generation
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Export

html-to-image

or

Playwright screenshots

Formats:

- PNG
 - PDF
-

14. Recommended MVP Scope

Build only:

- URL input
- Device library
- Portrait/landscape toggle
- Infinite canvas
- Linked navigation
- Linked scrolling
- Mockup system
- PNG export

Do NOT build:

- Authentication
- Teams
- Collaboration
- Comments

- Browser matrix testing
- AI QA
- Cloud sync

Focus on proving one hypothesis:

Designers and developers want a Figma-like workspace that allows them to review live websites across multiple responsive breakpoints simultaneously while preserving accurate viewport behavior.